

> AI coding agents wrote this robot arm's program. It packed groceries right 28 times in 30

UC Berkeley, NVIDIA and Bosch's GaP writes the program for about $0.01 of AI calls. Where does it still fall short of a factory?

> 01 — WHAT HAPPENED

GaP lets AI agents program a robot arm for each new task



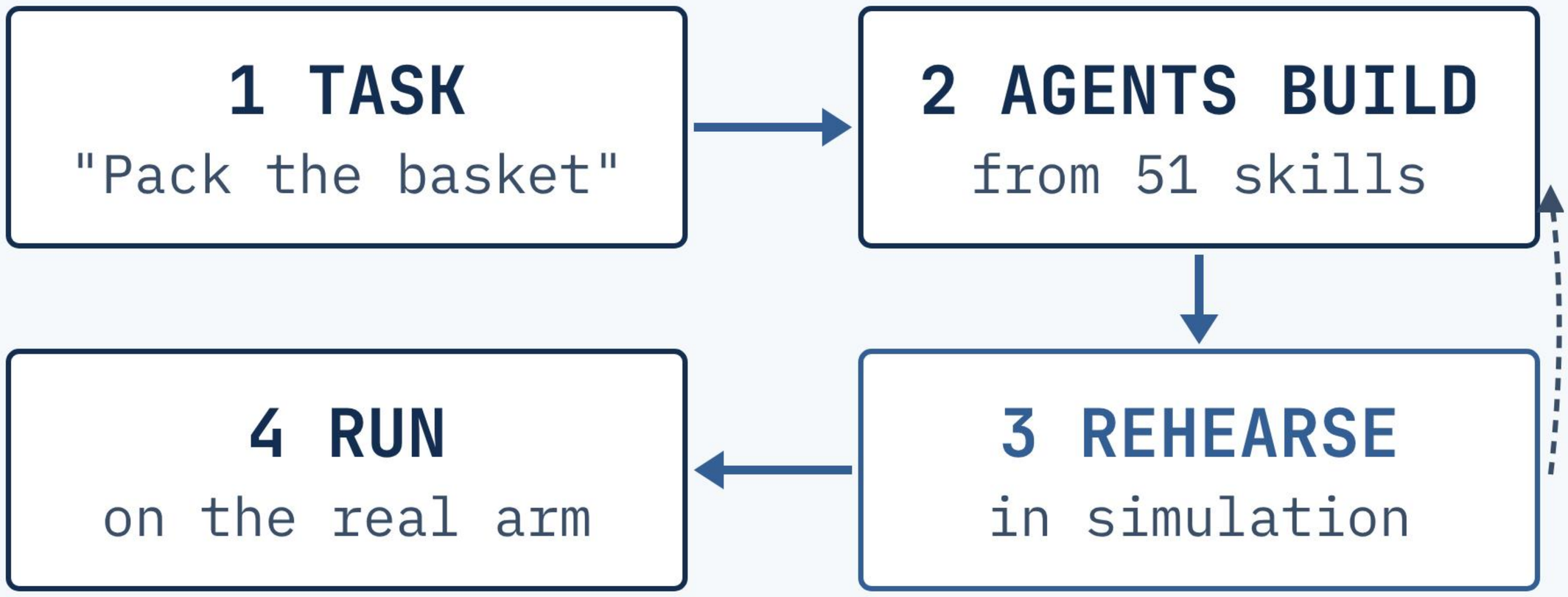
GaP, a spotlight paper at CoRL 2026 (the Conference on Robot Learning), doesn't train a new model. AI coding agents, language models that write and fix code, write the robot's program.

It targets factory jobs where the task repeats but parts vary in shape and position. Berkeley's Ken Goldberg says copying human demonstrations has plateaued there.

> 02 — HOW IT WORKS_

GaP turns a typed task into a flowchart of tested skills

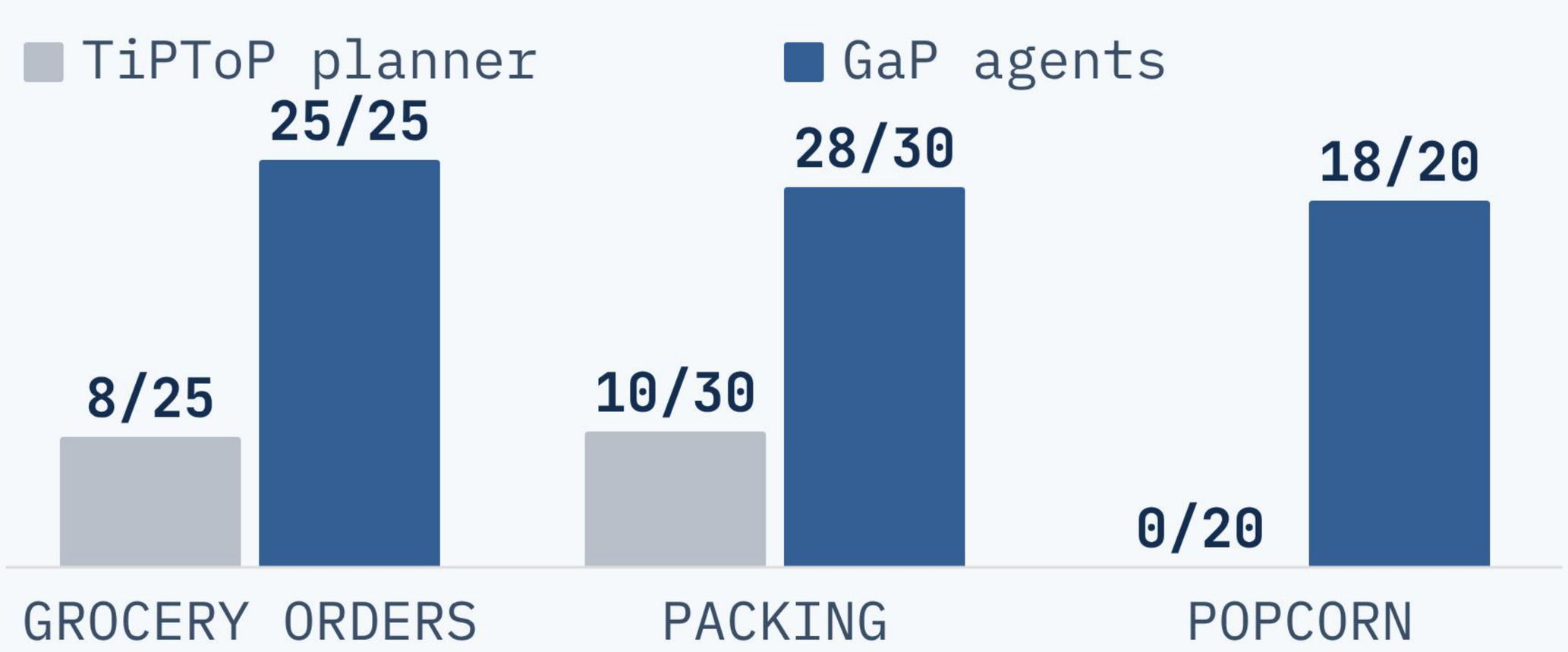
One agent splits the task into steps. Others fill each step from MORSL, a library of 51 existing tools: object finders, grasp and motion planners. The flowchart is rehearsed in NVIDIA Isaac simulation, and agents patch whichever box failed (dashed arrow). Swap the flowchart for one LLM writing plain Python and success drops to zero.



> 03 — THE RESULTS_

On a real Franka arm, GaP beat the TiPToP planner on 3 tasks

TiPToP, the grey bars, is a planner that chains similar perception and grasp tools, without a rehearsal step. For popcorn, rehearsal lifted simulated success from 33% to 94% in 10 rounds, about 150 minutes, the length of a long movie. The agents run on Gemini 3.1 Flash-Lite, a small Google model.



> 04 — THE CATCH_

GaP still runs about 10 times slower than a factory needs

The paper puts the industry bar at 7 seconds per item, about 500 an hour. GaP took 67 seconds per pick-and-place. Washing crates with two arms, it succeeded 143 times in 150, against 148 for a program hand-written by an expert.

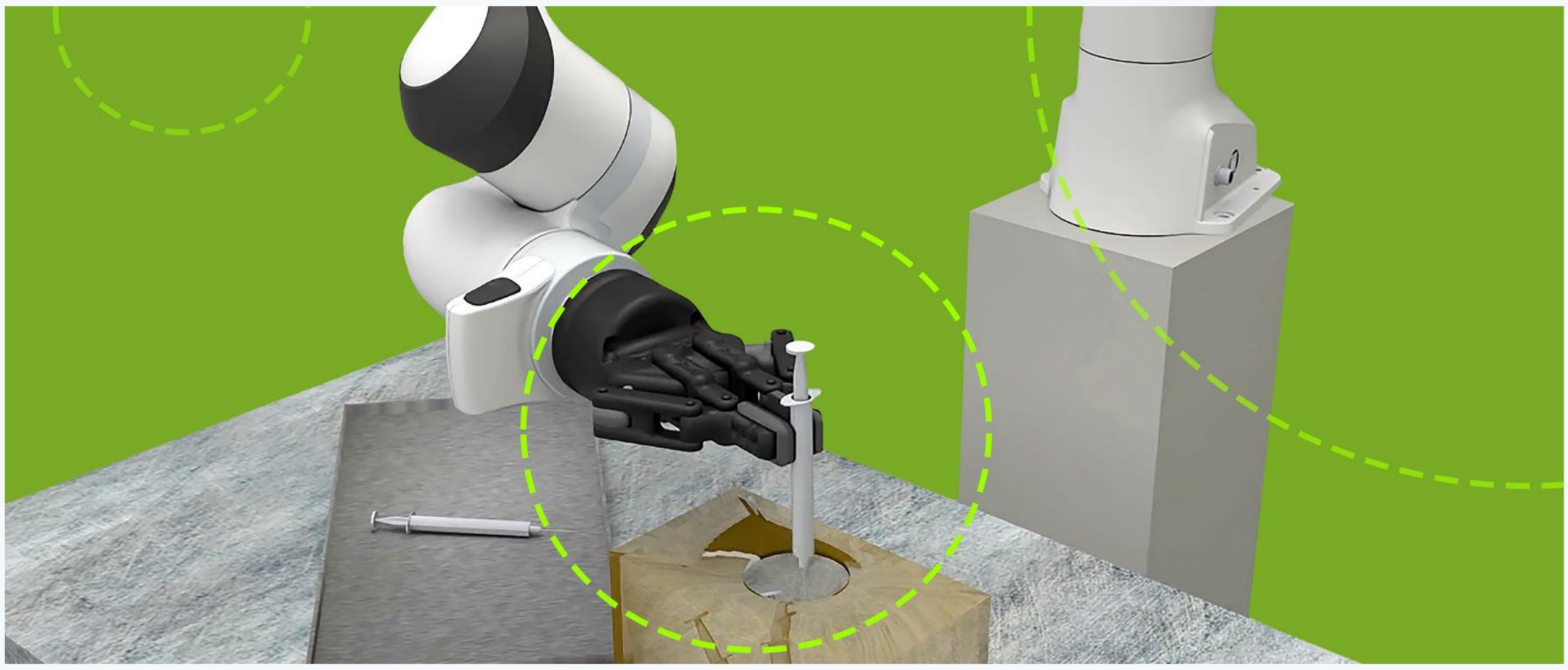
Every test is slow pick-and-place. Soft objects and moving targets are left for later.

"Execution reliability is not yet at industry levels."

— CHEN, XIE ET AL., GAP PAPER, ARXIV, NOV. 2026

> 05 — WHAT TO WATCH_

Berkeley and NVIDIA's VABAR tests agent-built robots



VABAR is an open benchmark of six industrial tasks with difficulty tiers, like this simulated sharps-disposal job. The same day, NVIDIA launched GR00T 2, trained partly on 100,000 hours of first-person human video. Watch whether other labs post VABAR scores, and whether GaP's 67 seconds per pick comes down.

> END OF TRANSMISSION_



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